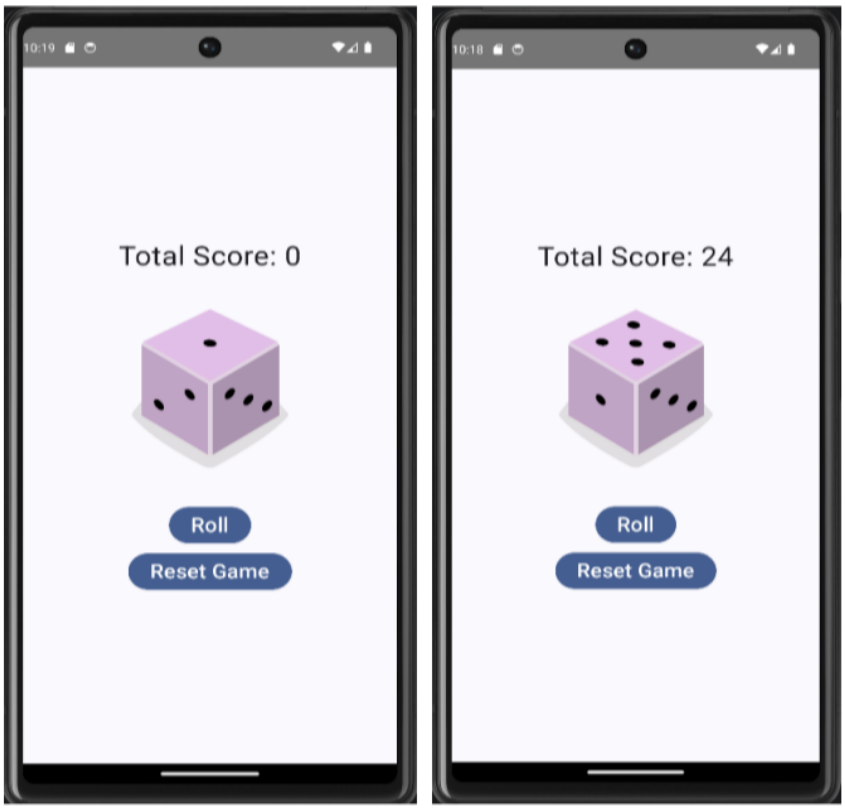
Task 2 — Designing Interactive Screen with Multiple Buttons & States

1. Feature Idea: Dice Roll & Total Score Tracker

This feature allows a user to roll a die to accumulate points.
It includes two primary interactions:

- **Roll Button**: Generates a random number and adds it to the **Total Score**.
- **Reset Button**: Clears the current score back to zero.

2. Screen Layout Sketch (Description)



Top Section

A large **Text** element displaying the Current Roll.
Example:
> "You rolled a 4!"

Middle Section

A **Text** element showing the cumulative total score.
Example:
> "Total Score: 15"

Bottom Section

A **Row** containing two buttons:

- **Button A:** "Roll Die" (Primary Action)
- **Button B:** "Reset Score" (Secondary Action, styled differently)

3. Button Actions & State Changes

Button A — Roll Die → State Change A

Logic: - Triggers `(1..6).random()` - Result is saved to `currentRoll` - Value is added to `totalScore`

UI Feedback: - Dice image/text updates to new number - Total Score counter increments immediately

Button B — Reset Score → State Change B

Logic: - Sets both `currentRoll` and `totalScore` back to `0`

UI Feedback: - Score display resets to `"0"` - Status message `"Score Reset!"` appears briefly

4. State Table

State Number	Trigger	Resulting UI Change
1	Initial App Launch	Dice shows "1", Total Score shows "0"
2	Click "Roll Die"	`totalScore` increases by dice value; Dice updates
3	Click "Reset Score"	`totalScore` returns to "0"; Status updates to "Ready"

5. Code Repository

GitHub Link:
app link

6. Reflection

If state is not managed properly, the UI can become inconsistent. For example:

- Total score updating while the dice image does not
- Stale values appearing after a reset

To ensure reliability:

- Use **Layout Inspector** to verify recomposition
- Rotate the device to test orientation changes
- Use `rememberSaveable` to preserve state
- Test across different emulator screen sizes
- Ensure buttons wrap properly and do not overlap